# Yuanshan Zhong

University at Buffalo | Biomedical Engineering

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**Research Interests**
AI-driven neuroengineering integrating neural signal analysis, representation learning, brain–computer interfaces, computational neurostimulation, and adaptive closed-loop neural systems.

## EDUCATION

**University at Buffalo, SUNY** *Aug 2025 – Present*

M.S. in Biomedical Engineering

**University of Science and Technology of China (USTC)** *Sep 2018 – Jun 2023*

B.S. in Computer Science

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## RESEARCH EXPERIENCE

**Research Assistant**
Wenyao Xu Lab, University at Buffalo

### Multimodal Neural Signal Analysis for Objective Biomarker Discovery in Auditory Disorders

University at Buffalo | Advisor: Prof. Wenyao Xu | Oct 2025 – Present

• Contributed to the SYNAPSE multimodal neurophysiological platform integrating around-ear EEG and pupillometry for objective assessment of auditory disorders, including tinnitus, hyperacusis, and misophonia.

• Developed EEG analysis pipelines incorporating signal preprocessing, ERP and spectral feature extraction, cross-channel synchronization analysis, and statistical testing; evaluated conventional machine-learning and deep-learning models (EEGNet/EEGNex) for disease-related neural pattern classification and characterization.

• Developing a self-supervised pretraining framework for around-ear EEG by aggregating multiple public datasets and adapting foundation-model representation learning to sparse, non-standard electrode layouts, with the goal of learning transferable neural representations across subjects, datasets, and downstream tasks.

• Applied pretrained EEG foundation models, primarily LaBraM, across electrode channels and auditory tasks to identify latent neural features associated with four auditory conditions, systematically evaluating disease-related representations across channels, temporal windows, and embedding dimensions.

• Investigated what information the foundation model encodes through layer-wise representation analysis, frequency/spatial/temporal feature decomposition, and controlled statistical experiments, tracing how disease-related signals emerge and transform across model layers and distinguishing robust neural structure from potentially confounded representations.

### Electrothermal Modeling of High-Density Electrode Grids for Transcutaneous Neurostimulation

University at Buffalo | Advisor: Prof. Filip Stefanovic | Aug 2025 – Present

• Developed coupled electrical–thermal finite-element models in COMSOL Multiphysics to simulate current distribution, Joule heating, and transient temperature responses in multilayer biological tissue during peripheral transcutaneous neurostimulation.

• Systematically evaluated conventional single electrodes and HD-inspired 1–16 electrode grid configurations across 128 combinations of stimulation current, pulse width, and frequency to characterize current-density distribution and tissue heating.

• Characterized how electrode subdivision, active electrode count, and total active area shape current concentration and thermal accumulation, identifying distinct electrothermal regimes and a nonlinear decline in stimulation-side thermal load with increasing electrode count.

• Performed area-matched and count-matched control analyses, mesh-convergence and numerical robustness evaluations, and parameter-space characterization to validate the computational framework and separate geometry-dependent electrothermal effects.

### AI-Optimized High-Density Electrode Arrays for Selective Muscle Stimulation

University at Buffalo | Advisor: Prof. Filip Stefanovic | Dec 2025 – Present

• Developing a computational framework combining finite-element modeling, numerical optimization, and machine learning to optimize current steering across high-density surface electrode arrays for selective activation of targeted muscle regions.

• Constructing an 8×8 independently controlled electrode-array FEM model with anisotropic muscle conductivity, using precomputed basis fields and linear superposition to efficiently evaluate arbitrary 64-channel current distributions.

• Building a superposition-based optimization pipeline to generate optimal electrode-current distributions across target locations, depths, and muscle-fiber orientations, and training inverse ML surrogate models for rapid prediction of stimulation configurations.

• Extending the framework toward anatomically informed stimulation models by incorporating heterogeneous tissue properties, realistic muscle geometry and fiber architecture, with the long-term goal of adapting a standardized optimization framework to specific body regions.

### Hybrid Learning for Robust Neural Control in Brain–Computer Interfaces

Independent Research | Apr 2025 – Present

• Developed an EEG-driven closed-loop neural control framework for 2D cursor navigation using BCI Competition IV-2a, integrating neural decoding with sequential decision-making in an interactive control environment.

• Developed a hybrid supervised–reinforcement learning architecture that compresses EEG features into structured low-dimensional control states and jointly integrates neural representations, decoder outputs, and task geometry for policy learning.

• Evaluated the framework under strict cross-subject ID/OOD splits and EEG perturbations, demonstrating improved convergence, training stability, cross-subject generalization, and robustness of compact neural control representations.

• Extending the framework toward adaptive closed-loop neural control with continuously updated EEG representations, time-varying task states and goals, and feedback-driven policy adaptation in dynamic environments.

### Event–Frame Hybrid Neural Network for Occlusion Removal

University of Science and Technology of China | Advisor: Prof. Xueyang Fu | Dec 2022 – Jul 2023

• Developed a multimodal framework integrating event streams and image frames.

• Designed a hybrid neural network combining spiking neural networks (SNN) and convolutional networks.

• Developed cross-modal feature fusion to improve reconstruction under occlusion conditions.

• Constructed a dataset with 900+ samples and evaluated reconstruction performance using PSNR and SSIM.

### Knowledge-Graph-Based Recommendation with Graph Representation Learning

Research Project | University of Science and Technology of China | 2022

• Studied the integration of knowledge graphs and collaborative filtering for explainable movie recommendation.

• Built a knowledge graph from Freebase entities linked to Douban movies and extracted multi-hop relational structures for representation learning.

• Implemented knowledge graph embedding models (TransE, TransR) and integrated them with matrix factorization to enhance item embeddings.

• Evaluated recommendation performance using ranking metrics (Recall@5/10, NDCG@5/10).

### Programming Language Extension and Compiler Implementation

Research Project | University of Science and Technology of China | 2021

• Implemented a YACC-based parser to resolve ambiguity in relational expression grammars and enable correct evaluation of chained comparison expressions.

• Extended the educational PL/0 compiler with multi-dimensional array support, including symbol-table representation and runtime memory layout computation.

• Developed compiler backend mechanisms for array indexing, assignment, and control flow (for-loops and logical operators) through stack-machine instruction generation.

## PUBLICATIONS & MANUSCRIPTS

**Zhong, Y., & Stefanovic, F. (2026).** Comparative electrothermal analysis of single and small-electrode grid configurations for HD-inspired peripheral transcutaneous neurostimulation. Frontiers in Bioengineering and Biotechnology. doi:10.3389/fbioe.2026.1904885

**Zhong, Y.*, Das, A.*, Rivera Rosario, E., Sun, W., & Xu, W**. (2026). From decoding performance to interpretation in EEG foundation-model representations. **Manuscript in preparation (submission planned for November 2026).**
*Co-first authors.

**Das, A., Rivera Rosario, E., Zhong, Y., Bo, W., Gherardi, A., Sun, W., & Xu, W. (2026).** Wearable ear-EEG can record brain activity during auditory tasks in hyperacusis. **Manuscript submitted to Scientific Reports.**

**Zhong, Y., & Stefanovic, F. (2026).** AI-optimized high-density electrode arrays for selective muscle stimulation. **Manuscript in preparation.**

## TECHNICAL SKILLS

**Neuroengineering**
EEG signal processing, neural representation learning & analysis, EEG foundation models, ERP analysis, time-frequency analysis

**Machine Learning**
PyTorch, deep learning, self-supervised learning, reinforcement learning, model interpretability

**Modeling & Simulation**
COMSOL Multiphysics, finite-element modeling, electrothermal modeling, bioelectric field simulation, numerical optimization

**Programming**
Python, MATLAB

**Scientific Computing & Data Analysis**
NumPy, SciPy, scikit-learn, MNE-Python

## HONORS & AWARDS

National Scholarship (Third Prize), University of Science and Technology of China, 2018
Third Prize, Provincial Level Physics High School Competition (China)
First Prize, City-Level Mathematics High School Competition (China)